

BIG DATA ANALYTICS (BITE413L)

"GraphGuard: Graph Neural Network-Based Anomaly Detection in Cryptocurrency Transaction Networks with LLM-Driven Explainability"

Project Review 1

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ABSTRACT

Cryptocurrency transaction networks are inherently graph-structured, with wallets as nodes and transfers as edges, making rule-based fraud detection ill-suited to capturing their complexity. This project proposes GraphGuard, a system that models blockchain transaction data as a dynamic graph and applies Graph Neural Networks (GNNs) to detect anomalous patterns such as unusual fan-out structures, rapid fund cycling, and community-bridging transactions, indicative of fraud, money laundering, or wash trading.

Node embeddings are generated using GraphSAGE and GCN architectures trained on historical transaction data, capturing both structural and temporal transaction behavior. To address the black-box nature of GNN predictions, the system integrates a Large Language Model (LLM) layer that converts flagged subgraphs and their risk scores into natural-language explanations, using retrieval-augmented generation (RAG) to ground these explanations in known fraud typologies and regulatory definitions.

The system is evaluated on public blockchain datasets (e.g., the Elliptic dataset) using precision, recall, and F1-score against labeled illicit transactions, with a qualitative assessment of explanation quality. This work shows how combining graph analytics with LLM-based explainability can produce fraud detection systems that are both accurate and auditable, a critical requirement for real-world financial compliance.

1. TITLE FINALIZATION

Project Title: GraphGuard – Graph Neural Network-Based Anomaly Detection in Cryptocurrency Transaction Networks with LLM-Driven Explainability

Justification of Title:

- GraphGuard: A memorable system identity that signals the dual role of the platform — modeling transaction graphs and guarding against illicit financial activity.
- Graph Neural Network-Based: Reflects the core methodological choice of representing wallets and transfers as a graph and learning over that structure, rather than treating transactions as independent, tabular records.
- Anomaly Detection: Emphasizes that the system is designed to surface statistically and structurally unusual behavior — fan-outs, cycling, community-bridging — rather than only matching known fraud signatures.
- Cryptocurrency Transaction Networks: Scopes the domain explicitly to blockchain-based financial graphs, distinguishing this work from generic network fraud detection in banking or telecom.

- LLM-Driven Explainability: Highlights the second pillar of the system — translating opaque GNN risk scores into auditable, human-readable justifications grounded in regulatory and typological knowledge.

2. INTRODUCTION AND MOTIVATION

2.1 Problem Context

Blockchain-based cryptocurrencies have enabled a new class of financial activity that is simultaneously transparent (all transactions are publicly recorded on the ledger) and pseudonymous (wallet addresses are not directly tied to real-world identities). This combination has made cryptocurrency networks an attractive vector for money laundering, mixing services, wash trading, ransomware payouts, and darknet-market settlements. Regulatory bodies and exchanges are under increasing pressure to identify illicit activity before funds are cashed out into fiat currency.

Traditional fraud detection approaches — rule-based thresholds on transaction amount, velocity, or counterparty blacklists — are fundamentally limited because they treat each transaction as an isolated event. In reality, illicit actors structure their activity across multiple hops, splitting and recombining funds through intermediary wallets specifically to evade simple threshold-based rules. Cryptocurrency transaction data is naturally graph-structured: wallets are nodes, transfers are directed, timestamped, weighted edges, and illicit behavior manifests as structural patterns — unusually high fan-out from a single source, rapid cyclical movement of funds designed to obscure origin, or edges that bridge otherwise disconnected communities. Detecting these patterns requires models that reason over the graph topology itself, not just per-transaction features.

A further obstacle to real-world adoption is interpretability. Even where machine learning models achieve strong detection accuracy, compliance teams, auditors, and regulators require a justification for why a wallet or transaction was flagged. A prediction alone — a risk score with no accompanying reasoning — is insufficient for actions such as freezing an account or filing a Suspicious Activity Report. This creates a central problem: the absence of a system that jointly (a) models transaction data at the graph level to capture structural fraud signals and (b) produces regulator-legible, typology-grounded explanations for its flags.

2.2 Research Opportunity

Graph Neural Networks have demonstrated strong performance on node and subgraph classification tasks across domains such as social network analysis, recommendation systems, and citation networks, by learning representations that aggregate information from a node's local neighborhood. Architectures such as GCN (Graph Convolutional Networks) and GraphSAGE

extend this to inductive settings, allowing models to generalize to previously unseen wallets — an essential property given that new addresses are created continuously on public blockchains.

In parallel, Large Language Models have shown the ability to synthesize structured, technical inputs into coherent natural-language narratives, and retrieval-augmented generation (RAG) allows such narratives to be grounded in external, verifiable knowledge sources rather than relying purely on the model's parametric memory. This creates a clear research opportunity: pairing a GNN's structural anomaly detection with an LLM-RAG explanation layer that retrieves relevant fraud typologies (e.g., FATF red-flag indicators, known laundering typologies such as layering and peel chains) and regulatory definitions to produce explanations that are both accurate and auditable. While GNN-based fraud detection and LLM-based explanation generation have each been studied independently, their combination — specifically for cryptocurrency transaction graphs — remains underexplored.

2.3 Motivation Summary

This project is motivated by the dual need for accuracy and auditability in cryptocurrency fraud detection. The objective is to build a system that detects structurally anomalous transaction patterns with high precision and recall, while simultaneously producing explanations that a compliance analyst can act on — bridging the gap between raw graph-level model outputs and the regulatory requirement for justified, evidence-backed decisions.

3. LITERATURE SURVEY

3.1 Overview

Research in graph-based fraud detection has expanded significantly since Weber et al. (2019) released the labeled Elliptic Bitcoin transaction dataset, establishing the first supervised benchmark for illicit-transaction classification with GNNs. Subsequent work has extended this line along several fronts: self-supervised embeddings that reduce dependence on scarce illicit labels (Inspection-L, 2023), richer subgraph-level benchmarks such as Elliptic2 (Bellei et al., 2024), efficient graph-transformer architectures for financial graphs (FraudGT, 2024), real-time detection on continuously evolving graphs (Spade, 2024), large-scale synthetic AML datasets to supplement Elliptic's limited scale (Altman et al., 2024), and unified frameworks that jointly detect anomalies across node, edge, and subgraph levels (UniGAD, 2024). Separately, wash-trading-specific detection has progressed through network-clustering and graph-cycle-detection methods evaluated on real exchange data.

A parallel and largely separate body of research addresses explainability in graph learning. Work evaluated directly on Elliptic (2025) has shown that extreme class imbalance destabilizes both learning and SHAP-based feature attribution, while newer LLM-based approaches (He et al., 2024; “From Nodes to Narratives,” 2025) demonstrate that Large Language Models can narrate GNN predictions in natural language, but so far only on molecular graphs or domain-agnostic settings, and without grounding in external, verifiable regulatory sources — a role that retrieval-augmented generation (Gao et al., 2023) is well suited to but has not yet been applied to financial-graph explanations.

While these works individually demonstrate strong detection performance or strong explanation quality, no system yet integrates both within a single, cryptocurrency-specific pipeline. GNN-based fraud detection papers report precision/recall/F1 metrics without addressing how a compliance analyst would interpret or act on a flagged subgraph. Conversely, LLM-based graph-explanation research is evaluated on non-financial benchmarks and rarely grounds explanations in domain-specific regulatory typologies such as the FATF red-flag indicators that compliance teams already reference. This gap directly motivates GraphGuard's combined architecture.

Sr. No.	Paper Title, Authors & Year	Techniques Used	Contribution / Focus	Challenge / Limitation
1	Weber et al., “Anti-Money Laundering in Bitcoin: Experimenting with GCNs for Financial Forensics” (2019)	GCN, Random Forest	Introduced the Elliptic labeled Bitcoin transaction dataset and the first GNN baseline for illicit-transaction classification	GCN underperformed Random Forest on this benchmark; no temporal modeling or explanation layer
2	Lo et al., “Inspection-L: Self-	Self-supervised	Learned wallet	Embeddings remain a

Sr. No.	Paper Title, Authors & Year	Techniques Used	Contribution / Focus	Challenge / Limitation
	Supervised GNN Node Embeddings for Money Laundering Detection in Bitcoin," Applied Intelligence (2023)	GraphSAGE-style pretraining	embeddings without heavy reliance on scarce illicit labels	black box; no mechanism to justify a flagged wallet to a reviewer
3	Bellei et al., "The Shape of Money Laundering: Subgraph Representation Learning on the Blockchain with the Elliptic2 Dataset," KDD MLF (2024)	Subgraph representation learning, GNN	Released Elliptic2 with community-level illicit/licit subgraph labels for richer AML benchmarking	Focused purely on subgraph classification accuracy; no explanation generation for flags
4	Lin et al., "FraudGT: A Simple, Effective, and Efficient Graph Transformer for Financial Fraud Detection," ICAIF (2024)	Graph Transformer, edge-attribute attention bias	Achieved 7.8–17.8% higher F1 with greater throughput on financial transaction graphs	Optimized for speed and accuracy; attention weights are not converted into reviewer-facing rationale
5	Jiang et al., "Spade: A Real-Time Fraud Detection Framework on Evolving Graphs," PVLDB (2024)	Streaming graph updates, incremental scoring	Enabled low-latency fraud scoring on continuously evolving transaction graphs	Prioritizes real-time throughput; does not address why a specific transaction was scored high-risk
6	Altman et al., "Realistic Synthetic Financial Transactions for Anti-Money Laundering Models," NeurIPS Datasets & Benchmarks (2024)	Multi-agent synthetic transaction simulation	Released a large, diverse synthetic AML benchmark to complement Elliptic's limited scale/label coverage	Synthetic patterns may not fully capture real-world adversarial laundering behavior
7	"An Explainable Graph Neural Network Framework for Anti-Money Laundering in Cryptocurrency Transactions Using the Elliptic Dataset" (2025)	GCN + SHAP under class imbalance	Examined how extreme class imbalance (<2% illicit) destabilizes GNN learning on Elliptic and applied SHAP for feature attribution	Explanations are feature-level SHAP values, not natural-language or typology-grounded narratives
8	Sirolly, Ma, Kanoria & Sethi, "Network-Based Detection of Wash Trading," Management Science / SSRN (2024–2025)	Iterative network clustering of colluding counterparties	Detected closed-cluster wash-trading rings on a real exchange (Polymarket) via network structure alone	Flags colluding clusters but does not classify individual transaction risk or explain the flag
9	"Wash Trading Detection Techniques for Centralised Cryptocurrency Exchange Services," AIBC (2024)	Graph-cycle detection (DFS on transaction graph)	Automated detection of collusive trading cliques via cycles in the transaction graph	Rule/topology-based rather than learned; no GNN representation or generated explanation
10	He et al., "Explaining Graph Neural Networks with Large Language Models: A Counterfactual Perspective for Molecular Property Prediction," EMNLP Findings (2024)	LLM-guided counterfactual graph explainer (LLM-GCE)	Used an LLM to generate counterfactual explanations for GNN predictions	Applied to molecular graphs; not grounded via retrieval in any regulatory/compliance knowledge base
11	"From Nodes to Narratives: Explaining Graph Neural Networks with LLMs and Graph Context" (2025)	LLM + graph-context prompting	Proposed a general framework for turning GNN outputs into natural-language narratives	Domain-agnostic; not evaluated on financial graphs and not grounded in external typology sources
12	Gao et al., "Retrieval-Augmented Generation for Large Language Models: A Survey" (2023)	RAG, dense retrieval	Established retrieval-grounding as a mechanism to reduce hallucination in knowledge-intensive LLM generation	General-purpose NLP survey; no application to graph-derived structured fraud evidence

Sr. No.	Paper Title, Authors & Year	Techniques Used	Contribution / Focus	Challenge / Limitation
13	FATF, "Virtual Assets and Virtual Asset Service Providers: Red Flag Indicators of Money Laundering and Terrorist Financing" (regulatory guidance)	Standardized typology cataloguing	Documents recognized laundering typologies (layering, structuring, mixing/peel chains) used by compliance teams	Not a technical system; used here purely as a grounding source, not integrated with any ML pipeline
14	Lin, Tang, Zi, Zhao, Yao & Li, "UniGAD: Unifying Multi-level Graph Anomaly Detection," NeurIPS (2024)	Unified node/edge/graph-level anomaly detection (MRQSampler)	Jointly modeled anomalies across node, edge, and subgraph levels in one framework, explicitly citing money-laundering as a motivating case	Evaluated on generic anomaly benchmarks, not cryptocurrency-specific; no explanation layer

3.2 Research Gaps Identified

Gap 1: No Unified Detection-Plus-Explanation Pipeline for Financial Graphs. Observation: Detection-focused studies such as Weber et al. (2019), Lo et al. (2023), Bellei et al. (2024), FraudGT (Lin et al., 2024), Spade (Jiang et al., 2024), Altman et al. (2024), and UniGAD (Lin et al., 2024) report detection performance but do not provide reviewer-facing explanations. LLM-based graph-explanation work such as He et al. (2024) and From Nodes to Narratives (2025) is not cryptocurrency-specific. Contribution: GraphGuard combines GNN risk scoring with an LLM-RAG explanation layer designed specifically for financial transaction graphs.

Gap 2: Explanations Are Not Grounded in Regulatory Typologies. Observation: The 2025 Elliptic explainability study uses SHAP feature attributions, while He et al. (2024) and From Nodes to Narratives (2025) generate natural-language explanations without grounding them in standardized regulatory typologies. Contribution: GraphGuard retrieves FATF red-flag indicators and fraud typology definitions so explanations use established compliance terminology such as "peel chain" and "layering".

Gap 3: Limited Joint Modeling of Temporal Cycling Alongside Learned Embeddings. Observation: Weber et al. (2019) and Lo et al. (2023) do not explicitly encode transaction timing or cyclical fund movement, while Spade (Jiang et al., 2024) focuses on streaming throughput rather than making cyclical patterns explicit and interpretable. Contribution: GraphGuard incorporates temporal edge features such as timestamp, recency, and velocity into GraphSAGE/ GCN neighbor aggregation so fund-cycling evidence can be both learned and explained.

Gap 4: Wash Trading and General AML/Fraud Detection Remain Separate Pipelines. Observation: Wash-trading studies such as Sirolly et al. (2024-2025) and the 2024 AIBC wash-trading study rely on network clustering or graph-cycle detection as separate pipelines from GNN-based AML systems. Contribution: GraphGuard unifies fan-out, fund cycling, community bridging, and wash-trading-style closed clusters within a single graph-representation-learning framework.

Gap 5: Class-Imbalance-Aware Explainability Has Not Been Extended to Natural Language. Observation: The 2025 Elliptic explainability study reports that extreme class imbalance, with illicit transactions below 2% of nodes, destabilizes both model learning and SHAP-based explanations, but it stops at feature attribution. Contribution: GraphGuard evaluates natural-language explanation quality, including accuracy, groundedness, and actionability, under the same imbalanced-label conditions.

4. PROBLEM STATEMENT

4.1 Formal Problem Definition

Existing cryptocurrency fraud detection approaches either (a) rely on per-transaction, rule-based heuristics that ignore the graph structure of wallet-to-wallet fund movement, or (b) apply graph-based machine learning models that achieve reasonable detection accuracy but produce opaque risk scores with no accompanying justification. Neither approach satisfies the joint requirement of real-world financial compliance: structurally-aware detection of illicit patterns and human-legible, regulation-grounded explanations for every flagged wallet or subgraph.

4.2 Research Gap Definition

There exists no integrated system that simultaneously (1) represents cryptocurrency transaction data as a dynamic, temporally-aware graph, (2) applies inductive Graph Neural Network architectures (GraphSAGE, GCN) to classify wallets and subgraphs as illicit or licit with competitive precision/recall/F1 against labeled benchmarks such as Elliptic, and (3) converts flagged predictions into natural-language explanations grounded via retrieval-augmented generation in known fraud typologies and regulatory definitions. As the literature survey shows, detection-focused papers evaluated on Elliptic and related benchmarks (Weber et al., 2019; Lo et al., 2023; Bellei et al., 2024; Lin et al., 2024) report strong precision/recall/F1 but do not address explanation generation, while LLM-based graph-explanation methods (He et al., 2024; “From Nodes to Narratives,” 2025) are evaluated outside the financial-graph domain and are not grounded in regulatory typologies. This leaves a clear gap for a unified, auditable fraud-detection framework specific to cryptocurrency transaction networks.

5. PROJECT OBJECTIVES

5.1 Primary Objective

The primary objective of this project is to design, develop, and validate GraphGuard, a scalable big-data analytics and graph-learning system for cryptocurrency transaction networks. The system stores heterogeneous blockchain datasets in a data-lake architecture, uses Apache Spark/PySpark for large-scale preprocessing and feature engineering, constructs dynamic transaction graphs, applies Graph Neural Networks to detect anomalous wallets and subgraphs, and integrates an LLM-driven retrieval-augmented explanation layer so that detected activity can be reviewed and audited for financial-compliance use.

5.2 Specific Research Objectives

Big-data storage: Design a data-lake architecture for Elliptic/Elliptic2, supplementary public blockchain data, and synthetic AML datasets. The lake is organized into raw, processed, and analytics zones and can be backed by HDFS or S3-compatible object storage depending on deployment.

Distributed preprocessing: Use Apache Spark/PySpark for scalable ingestion, cleaning, schema validation, joins between transaction features, graph edges and labels, aggregation, and structural/temporal feature engineering before GNN training.

Graph construction: Construct a graph representation of blockchain transaction data, treating wallet addresses or transactions as nodes and transfers as directed, timestamped, weighted edges. Engineer features such as in-degree, out-degree, fan-out ratio, transaction velocity, recency, and cyclical-path indicators.

GNN modeling: Train and compare GraphSAGE and GCN architectures for node/subgraph classification on the Elliptic dataset and, where feasible, supplementary public blockchain data, producing per-wallet or per-transaction risk scores.

Anomaly categorization: Detect unusual fan-out structures, rapid fund cycling, and community-bridging transactions so that high-risk activity is categorized rather than only assigned an undifferentiated score.

Explainability: Build a retrieval-augmented explanation layer that takes flagged subgraphs and their risk scores, retrieves relevant fraud typologies and regulatory definitions such as FATF red-flag indicators, and prompts a Large Language Model to produce a grounded natural-language explanation referencing the specific structural evidence behind the flag.

Evaluation: Evaluate detection quantitatively using precision, recall and F1-score against Elliptic's labeled illicit/licit transactions, and evaluate generated explanations qualitatively for accuracy, grounding and actionability.

Modular backend: Expose the storage, Spark preprocessing, graph construction, GNN inference and LLM-RAG explanation components through modular services and RESTful APIs so the system can later be extended toward production deployment.

5.3 Proposed System Architecture

The revised GraphGuard architecture introduces an explicit big-data storage and processing layer before graph construction. Raw Elliptic/Elliptic2 records, public blockchain datasets and synthetic AML benchmark data are first stored in a data lake. The lake separates raw source data from cleaned/processed datasets and downstream analytics artefacts such as graph-ready feature tables, embeddings and risk outputs.

Apache Spark/PySpark performs distributed preprocessing over this storage layer. Its responsibilities include data cleaning, joining the Elliptic feature, edge-list and class datasets, large-scale aggregation, normalization, and generation of temporal and structural features required for graph learning. Spark is used for ETL and feature engineering; GraphSAGE/GCN remains responsible for graph-neural-network training and inference.

The processed data is converted into a dynamic transaction graph and passed to GraphSAGE and GCN models for embeddings and risk scoring. Flagged activity is categorized into fan-out, rapid fund cycling and community-bridging patterns. The flagged subgraph, risk score and supporting graph evidence then feed the existing RAG-LLM explanation layer, which retrieves relevant FATF typologies and generates a grounded explanation for the compliance dashboard.

Kafka is intentionally not included in the current implementation. The present evaluation is batch-oriented and uses public benchmark datasets rather than a live transaction stream. Real-time ingestion can be treated as a future extension if GraphGuard is later connected directly to continuously arriving blockchain transactions.

5.3.1 Big Data Technology Rationale

Data Lake: provides a single scalable repository for heterogeneous raw, processed and analytics datasets without forcing all data into one rigid schema.

Apache Spark: provides distributed, fault-tolerant processing for ETL, joins, aggregation and feature engineering as data volume grows beyond a single-machine workflow.

HDFS / S3-compatible storage: are deployment options for the data-lake layer; the architecture is storage-agnostic and does not require a Hadoop cluster for the initial prototype.

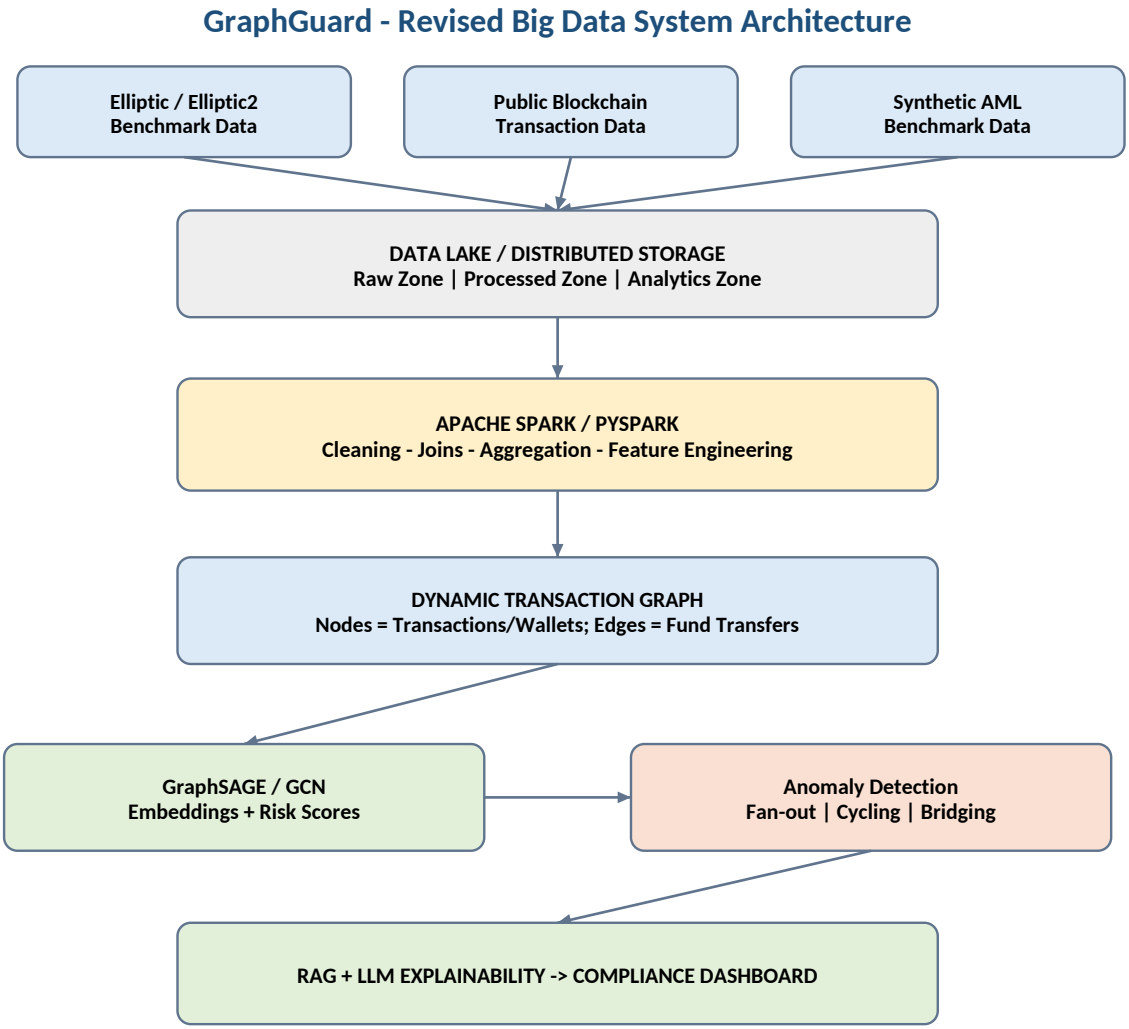


Figure 1 : Revised Architecture Diagram

The revised architecture makes the Big Data Analytics component explicit: datasets first pass through a data-lake storage layer and Apache Spark preprocessing before graph construction and GNN inference. This separation keeps distributed data engineering distinct from graph-model training while preserving the existing anomaly-detection and LLM-RAG explainability pipeline.

5.4 Summary of the Processing Flow

The revised processing flow begins with batch ingestion of Elliptic/Elliptic2, public blockchain and synthetic AML datasets into the raw zone of the data lake. Spark then validates schemas, cleans records, joins feature, edge and label data, performs aggregations, and computes structural and temporal attributes. The resulting graph-ready datasets are stored in processed/analytics zones before being converted into the transaction graph used by GraphSAGE and GCN.

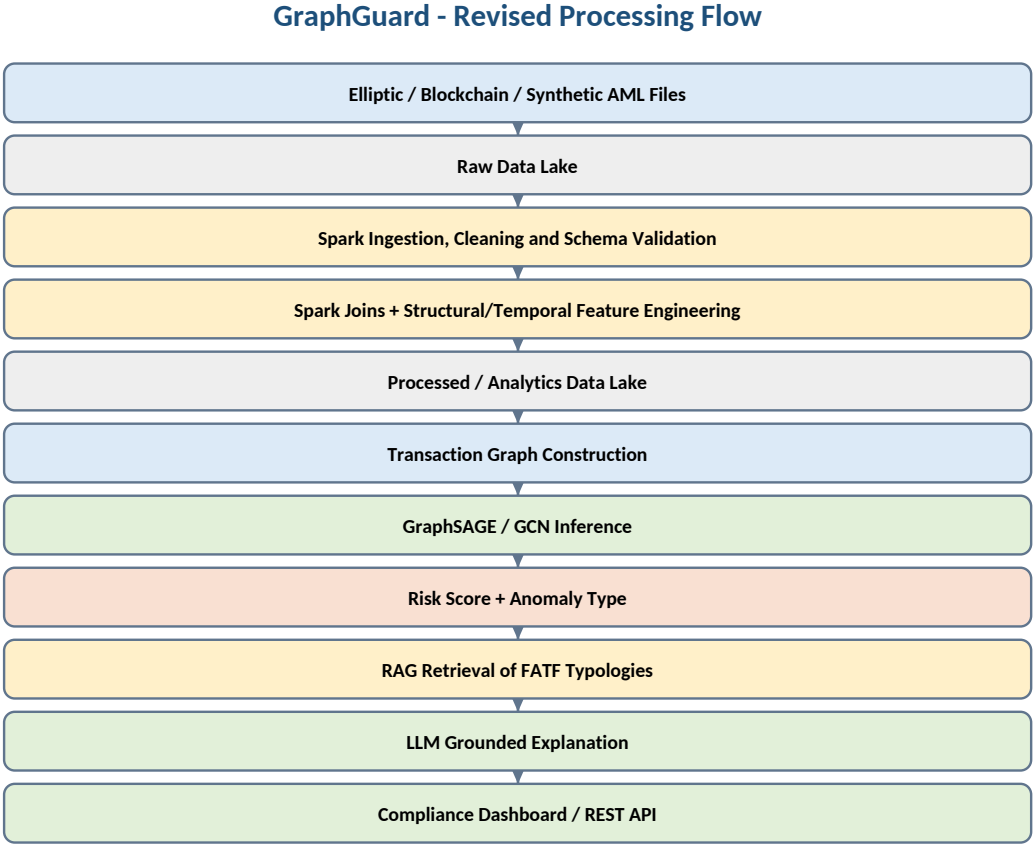


Figure 2 : Revised Processing Flow Diagram

6. EXPECTED OUTCOMES AND RESEARCH CONTRIBUTIONS

The primary expected outcome is a working big-data-to-GNN pipeline in which Apache Spark performs scalable preprocessing and feature engineering over transaction datasets stored in a data-lake architecture, after which GraphSAGE/GCN assigns interpretable risk scores to wallets and subgraphs within the Elliptic transaction graph. Detection performance will be evaluated using precision, recall and F1-score against published baselines on the same benchmark.

A second expected outcome is reliable categorization of flagged activity into the three targeted anomaly types - fan-out, fund cycling, and community-bridging - enabling more precise downstream triage than a single undifferentiated risk score would allow.

A third expected outcome is the generation of natural-language, typology-grounded explanations for flagged subgraphs, evaluated qualitatively for accuracy and groundedness, demonstrating that RAG-based grounding measurably reduces unsupported or hallucinated claims compared to ungrounded LLM prompting.

In terms of research contributions, the project introduces a graph-representation-learning framework tailored specifically to cryptocurrency transaction networks that treats fan-out, cycling, and community-bridging as distinct, jointly-modeled anomaly categories rather than a single fraud/non-fraud label.

A further contribution is the integration of retrieval-augmented LLM explanation generation with GNN-based detection within a single financial-fraud pipeline — an intersection that existing literature addresses only separately — demonstrating a concrete method for making graph-neural predictions auditable by grounding them in externally verifiable regulatory text.

Finally, the project contributes a modular backend design suitable for extension toward real-time transaction ingestion and integration into exchange or regulator-facing compliance tooling, positioning the work as both a practical prototype and a research contribution to explainable graph-based fraud detection.

7. CLARITY OF RESEARCH SCOPE

7.1 Scope Definition

The scope of this project is bounded to the analysis of cryptocurrency transaction graphs derived from publicly available blockchain data, with the Elliptic dataset as the primary evaluation benchmark and other public datasets considered as supplementary sources where feasible. The project focuses on wallet-level and subgraph-level anomaly detection — fan-out, fund cycling, and community-bridging patterns — rather than attempting to detect every conceivable category of financial crime.

The system encompasses graph construction and feature engineering, GNN model training and evaluation (GraphSAGE and GCN architectures), and an LLM-RAG explanation layer grounded in a curated corpus of fraud typologies and regulatory red-flag definitions. The scope explicitly includes quantitative evaluation (precision, recall, F1-score) and qualitative evaluation of explanation quality, and excludes building a production-grade, real-time exchange-integrated compliance product; real-time streaming ingestion is discussed as a future extension rather than an in-scope deliverable.

The project does not attempt to de-anonymize wallet owners or establish real-world identity behind addresses — it operates purely on graph-structural and behavioral signals available on-

chain. It also does not extend to non-cryptocurrency financial fraud domains (e.g., credit card fraud, insurance fraud); the framework is scoped specifically to blockchain transaction graphs. The explanation layer is scoped to generating human-readable justifications for already-flagged subgraphs, not to autonomously initiating regulatory action such as freezing funds or filing reports.

In summary, the scope is bounded by four principles: focus on graph-structural anomaly detection in cryptocurrency transaction networks, use of established public benchmarks (Elliptic) for quantitative validation, integration of retrieval-grounded LLM explainability as a first-class, formally evaluated system component, and design of a modular backend suitable for future extension rather than immediate production deployment.

7.2 Success Criteria

The success of the project is determined by measurable indicators across detection performance, explanation quality, and system design. Detection success is measured through precision, recall, and F1-score on the Elliptic dataset's labeled illicit/licit transactions, benchmarked against published GCN and GraphSAGE baselines.

Explanation quality is evaluated through a structured qualitative rubric assessing whether generated explanations (a) accurately reference the structural evidence underlying a flag, (b) are grounded in retrieved typology or regulatory text rather than unsupported LLM generation, and (c) would be actionable for a compliance analyst deciding whether to escalate a flagged wallet.

From a systems perspective, success is achieved if the graph-construction, GNN-inference, and explanation-generation components remain modular and independently testable, and if the exposed APIs return risk scores, anomaly categories, and explanations reliably for a given wallet or subgraph query. Academically, the project is considered successful if it demonstrably narrows the identified gap between graph-based fraud detection and typology-grounded explainability, providing a reusable baseline for future research in auditable financial graph learning.

8. CONCLUSION

This document has presented the conceptual and architectural foundation for GraphGuard, a system that combines Graph Neural Network-based anomaly detection with LLM-driven, retrieval-augmented explainability for cryptocurrency transaction networks. The study identified that current fraud-detection approaches either ignore the graph-structured nature of blockchain transactions or produce accurate-but-opaque risk scores unsuitable for regulatory use.

The literature survey demonstrated substantial progress in both graph-based fraud detection and LLM-based explanation generation individually, but revealed a clear absence of systems that

integrate the two specifically for cryptocurrency transaction graphs, and that ground explanations in standardized regulatory typologies. These gaps directly informed the project's objectives: constructing a temporally-aware transaction graph, training GraphSAGE and GCN models for anomaly classification across fan-out, cycling, and community-bridging patterns, and building a RAG-grounded LLM layer to translate flagged subgraphs into auditable, natural-language justifications.

The expected outcome is a modular, benchmarked prototype that achieves competitive detection performance on the Elliptic dataset while producing explanations that are both accurate and grounded, directly addressing the compliance requirement that flagged financial activity be explainable and auditable. Overall, this stage establishes the groundwork required to implement and evaluate a fraud-detection framework that is simultaneously structurally aware and interpretable — a combination that is critical for real-world adoption in financial compliance settings.